

Introduction to Artificial Intelligence

Project 22: Movie revenue prediction

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1 Project description

The goal of the project is to create a model for movie revenue prediction. Given a set of movie attributes (e.g., release date, budget, language, etc.) as input, the model should output a revenue prediction.

The data is available on Kaggle. The student should perform extensive data analysis and preprocessing, including visualizing, cleaning, analyzing, and splitting the data. The student may also apply data augmentation or find additional data if possible.

The choice of model is part of the task. The architecture should be selected carefully, taking into account factors such as the data, computational resources, and final performance.

The evaluation of the model should include commonly used metrics for the problem (e.g., test accuracy) as well as visualization of the outputs.

All important decisions made in the project should be properly justified and tested experimentally. For example, questions such as whether data augmentation helps or whether architecture/method A is better than B should be answered theoretically if possible and supported by experiments. An ablation study of the final solution is also highly valuable.

2 Schedule

The project is done in pairs and has three stages. Maximum number of points from the project is 25. The project timeline, with points for each stage, is as follows:

- Preliminary assumptions submission deadline is May 15th. The meeting will be arranged during the week starting May 18th. (0–5 points)
- The midterm solution should be delivered by the end of May 24th. The meeting will be arranged during the week starting May 25th. (0–5 points)
- The final solution delivery deadline is June 9th. The meeting will be arranged on per pair basis on 15th and 16th June. (0–15 points)

Delay in handing in each stage results in a penalty of 20% of total points available from the stage. **After each deadline, we will have project meetings on MS Teams.** The meetings are **mandatory** to complete the project, and we will do them in a week following the handing in of each stage at the lab timeslots. I will schedule the meetings. In case you are not available, please let me know, so we can find a new slot. In case of any questions about projects, either ask me directly on Teams.

3 Technical details

Solutions should be implemented in Python. Jupyter Notebooks are not allowed. Please ensure that your code adheres to basic standards of lean coding in accordance to PEP 8 and add instructions on how to run the code. The report should follow the structure of a scientific paper, preferably using a LaTeX template. Please remember to include tables with performance measures, plots, etc.; do not include code blocks in the report. The code, alongside the reports, should be uploaded on GitHub. Please send the links to the repository as submissions for each stage of the project.

4 Assessment criteria

Preliminary report

The preliminary report (short, 1 page) should be a PDF file containing:

- Description of the task that you are working on, e.g.: what is the goal of the project and how do you frame the task you want to solve (regression, binary classification, multilabel classification). (1 point)
- Description of the dataset. (1 point)
- Description of proposed solution (3 points): Describe briefly the usual methods used for the problem. Select two or three of them that you want to implement and describe them briefly.

Midterm solution

At this stage, you should present your progress, e.g. what have you already implemented and how it works and include its description in the extended report. The grading of this stage will take into account:

- Your progress in coding and evaluation of the method — one of the selected algorithms should be implemented at this stage in the preliminary form. (1 point)
- Intermediate results and their presentation. Please add the results to the project report. The results should be presented accordingly, as plots or tables. Make sure your results are reproducible by adding a seed to your code. (1 point)
- Presentation and discussion on the challenges, results, and findings during the project so far, and potential corrections to the previous assumptions. This should also be included in the report. (2 points)
- Discussion on the plans for finishing the project, e.g. what experimental results do you want to show in the final report? (1 point)

Final solution

The final solution should be the code and report. The grading of this part will be:

- Submission of the code for your solution and correctness of the code. (5 points)
- The final report should contain the information from the previous stages. The grading of the report will be as follows:
 - The proper form of the report follows the format of a scientific paper (introduction, description of the data and algorithms that you will use, intermediate results and insights from the midterm solution, final experimental results, and conclusions). Do not include the code in your report. (2 points)
 - Experiments, presentation (tables, plots), and the discussion of the results. (4 points)
 - Comparison of the algorithms that you implemented, based on the results from the experimental section. (2 points)

- Final conclusions — which algorithm performed the best, is the solution satisfactory and how could it potentially be improved? (2 points)

In case of any questions contact me via MS Teams.